Iniciación práctica al análisis de datos OMOP

Importancia de RWD para guiar esfuerzos en el ámbito regulador: el ejemplo de DARWIN EU®

25ANYS IDIAP JORDI GOL

By 2025 the use of Real-World Evidence will have been enabled and the value will have been established across the spectrum of regulatory use cases

- European Medicines Regulatory Network (EMRN) strategy to 2025 -

Classified as public by the European Medicines Agency

Three main areas for which RWD analyses can support regulatory decision-making

1 Support the planning and validity of applicant studies

- Design and feasibility of planned studies
- Representativeness and validity of completed studies

2 Understand the clinical context

- Disease epidemiology
- Clinical management
- Drug utilisation

3 Investigate associations and impact

- Effectiveness and safety studies
- Impact of regulatory actions

Classified as public by the European Medicines Agency

Towards delivering the 2025 RWE vision

EMA studies using in-house databases

- **Primary care** health records from the **France**, **Germany, UK, Italy, Spain** and **Romania.** Some data sources include data on specialist.

Studies procured through EMA FWCs

- New framework contract (FWC) since September 2021: services of **8 research organisations** and academic institutes
- Access to **wide network of data sources**: 59 data sources from 21 EU countries
- Ability to leverage external **scientific expertise**

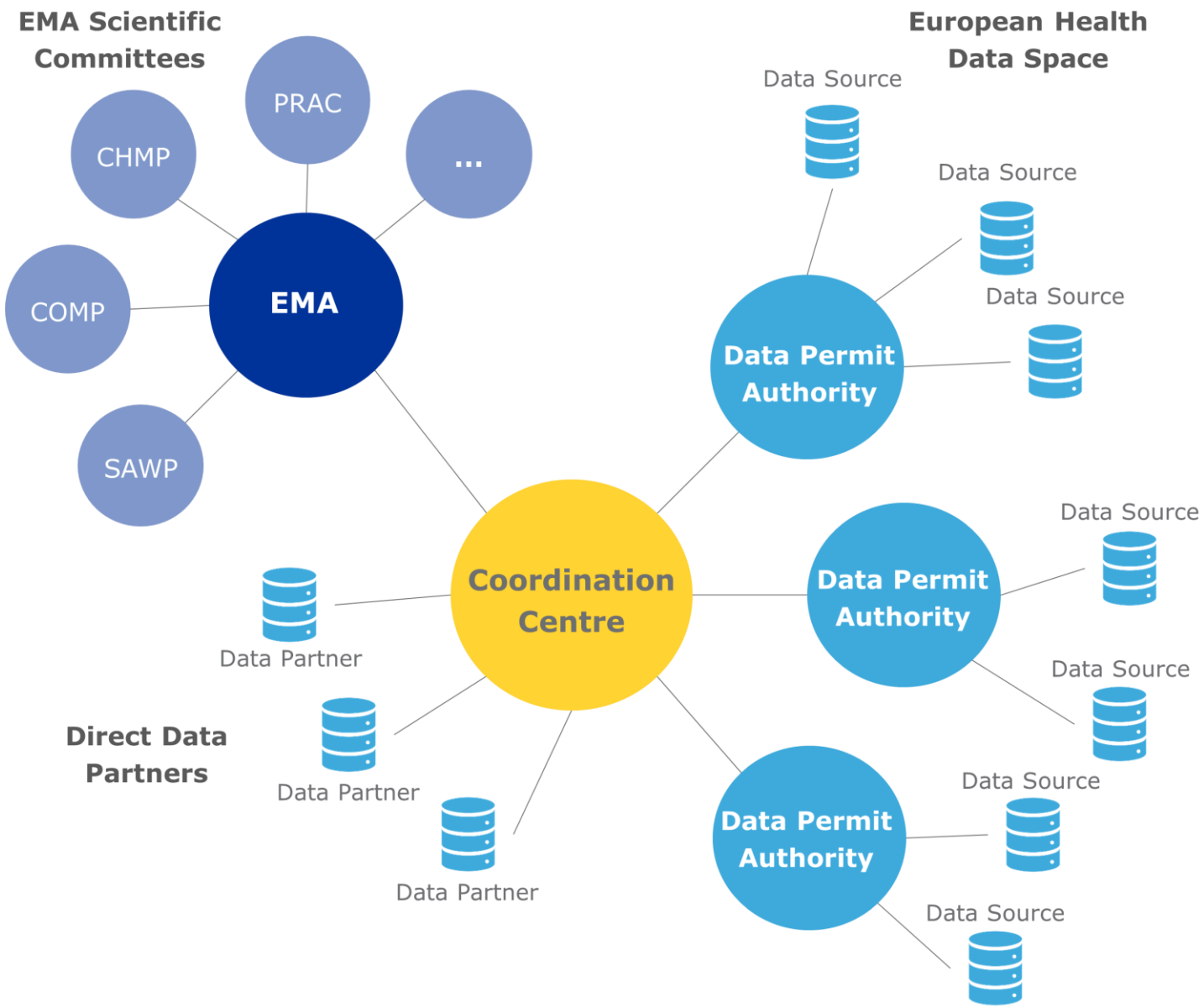
DARWIN EU®

- Coordination Centre launched February 2022
- Onboarded first **39 data sources**
- Additional 10 data partners are foreseen to **be added in 2025**

Classified as public by the European Medicines Agency

DARWIN EU® - Data Analysis and Real World Interrogation Network

DARWIN EU® is a federated **network** of **data**, **expertise** and **services** that supports better decision-making throughout the product lifecycle by generating reliable **evidence from real world healthcare data**



FEDERATED NETWORK PRINCIPLES

- Data stays **local**
- **Use of OMOP Common Data Model** (where applicable) to perform studies in a timely manner and increase consistency of results

DARWIN EU Network of Data Partners

International data platform
HARMONY Big Data Platform

The Netherlands
Integrated Primary Care Information
Netherlands Cancer Registry

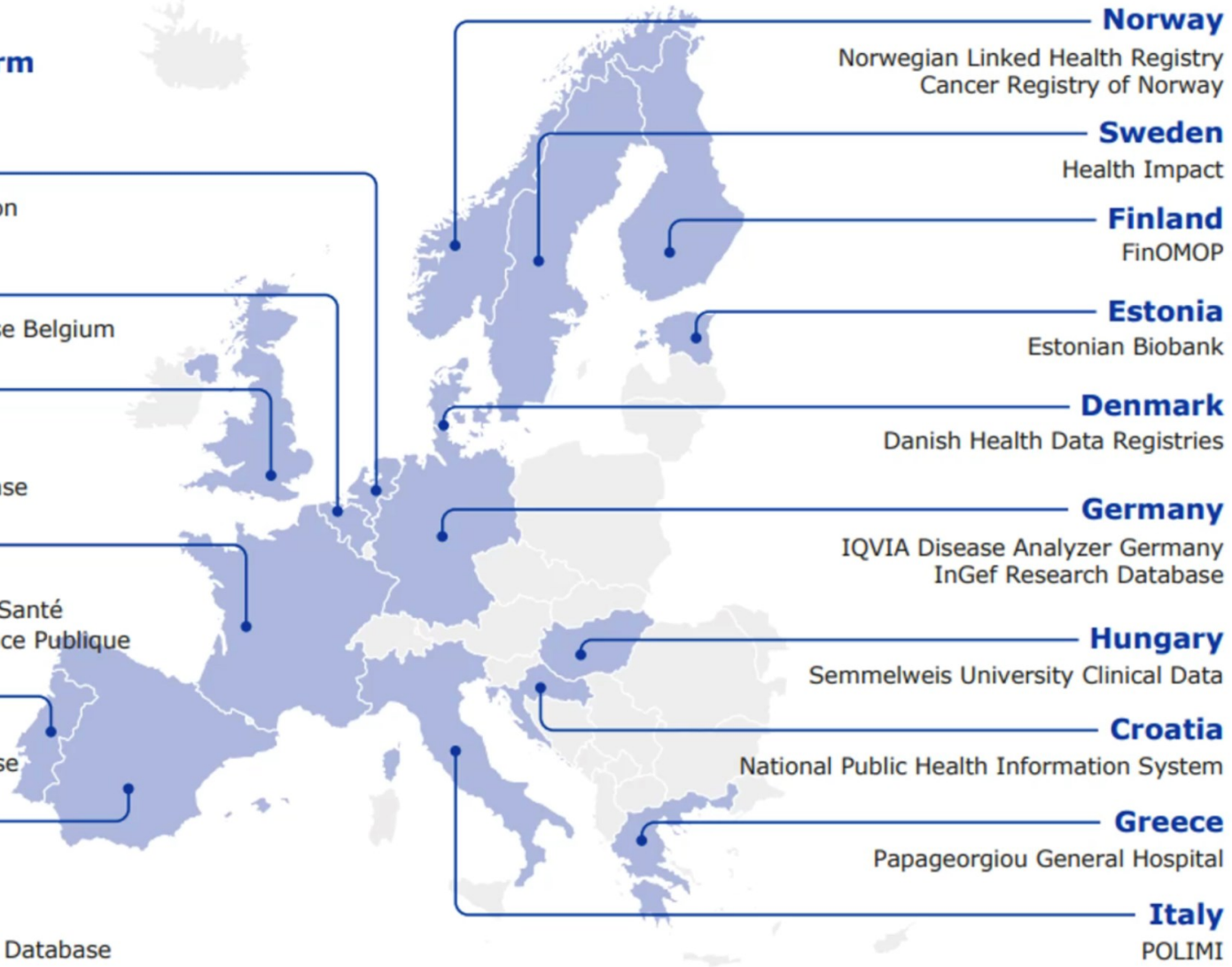
Belgium
IQVIA Longitudinal Patient Database Belgium

United Kingdom
UK BioBank
Clinical Practice Research Datalink
National Neonatal Research Database

France
Bordeaux University Hospital
Système National des Données de Santé
Health Data Warehouse of Assistance Publique

Portugal
ULSM-RT
Egas Moniz Health Alliance DataBase

Spain
SIDIAP
BIFAP
IMASIS and IMIM
Valencia Health System Integrated Database
H12O
Health Data Research Platform of the Balearic Islands



Norway
Norwegian Linked Health Registry
Cancer Registry of Norway

Sweden
Health Impact

Finland
FinOMOP

Estonia
Estonian Biobank

Denmark
Danish Health Data Registries

Germany
IQVIA Disease Analyzer Germany
InGef Research Database

Hungary
Semmelweis University Clinical Data

Croatia
National Public Health Information System

Greece
Papageorgiou General Hospital

Italy
POLIMI

30 Data Partners as of Feb 2025 (~ +10 by end of Feb 2026) in **16 European countries**

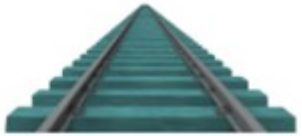
~100 studies per year from 2025 onwards

EMA

What is needed to facilitate observational studies at scale?

Data interoperability

Standardised analytics



Technical Infrastructure

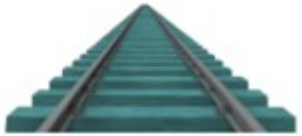
Data network

We need to make studies repeatable, reproducible, replicable, generalisable, and robust

What is needed to facilitate observational studies at scale?

Data interoperability

Standardised analytics



Technical Infrastructure

Data network

Generating Reliable Evidence using the OMOP Common Data Model

We need to make studies repeatable, reproducible, replicable, generalisable, and robust

What analyses and studies will DARWIN EU® deliver?

Category of observational analyses and studies	Description
Routine repeated analyses	**Routine analyses** based on a **generic study protocol** • Periodical estimation of drug utilisation • Safety monitoring of a medicinal product • Estimation of the incidence of a series of adverse events
Off-the-shelf studies	Studies for which a **generic protocol** is adapted to a research question • Estimate the prevalence, incidence or characteristics of exposures • Health outcomes • Describe population characteristics
Complex Studies	Studies **requiring development or customisation** of specific study designs, protocols • Etiological study measuring the strength and determinants of an association between an exposure and the occurrence of a health outcome considering sources of bias, potential confounding factors and effect modifiers
Very Complex Studies	Studies which **cannot rely only on electronic health care databases**, or which would require **complex methodological work** • Studies where it may be necessary to combine a diagnosis code with other data such as results of laboratory investigations, or studies requiring additional data collection

Number of studies

PHASE I Establishment – 1st year → PHASE II Establishment – 2nd year → PHASE III Operation – 1st year → Operation 2nd year → Operation 3rd year

	Year 1	Year 2	Year 3	Year 4	Year 5
Phases	Phase I	Phase II	Phase III	Option 1	Option 2
Routine Repeated analysis	-	-	4	35-50	35-50
Off the shelf studies	3	14	23	35-50	35-50
Complex Studies	1	4	10	15-20	15-20
Very Complex Studies	0	0	0	1	1

https://darwin-eu.org/index.php/studies

1. Study Exploration	<u>Study Feasibility- Study request by EMA:</u> - Do we have the data? – Darwin Portal - Do we have the analytical pipelines? (OTS)
2. Study Initiation	- Work Order Form Data Partners - Creation of Study Team: PI/data analyst - Declaration of Interest
3. Study Implementation	- Study outline/Protocol – Upload to EUPAS register - IRB approval - Kick-off meeting - Phenotyping – Cohort Diagnostics - Study Package – Test Run
4. Study Execution	- Data Partners run Study Package - Data QC by Data Partners - Results uploaded to DRE - Results reviewed by PI
5. Study Dissemination	- Generation of Study Report (ENCePP template) - Upload to EUPAS register - Manuscript generation - Study archiving

EMA

Database Partners

Darwin CC:
- Network Pillar
- Development Pillar
- Technology Pillar
- Management Pillar

Catalogue of Standard Analyses:
Off-the-shelf studies and examples

Standard Analysis	Regulatory example
Population-level disease epidemiology	• Prevalence of rare disease/s • Background rates of AESI
Patient-level disease epidemiology	• Natural history/prognosis • Current practice/treatment patterns
Population-level DUS	• Incidence and prevalence of use of medicine/s over time
Patient-level DUS	• Describing indication/s for drug/s • Treatment duration, cumulative use

Catalogue of Standard Analyses:
Complex studies and examples

Standard Analysis	Regulatory example
RMM Effectiveness	• Incidence of drug/s use before and after a regulatory action • Medicine/s user/s profile after new indication or contraindication
New user, active comparator, cohort studies	• Post-authorisation safety study • Comparative effectiveness
Self-controlled case series	• Vaccine safety surveillance

STUDIES YEAR 1

Study Title	Committees	Study Type	Type of analysis	Data bases
DARWIN EU® - **Prevalence of rare blood cancers in Europe**	COMP	OTS	Disease Epidemiology	IPCI (Nl) **SIDIAP (Spain)** CPRD Gold (UK) IQVIA LPD (Be) IQVIA DA (Ge)
DARWIN EU® - **Drug utilisation of valproate**-containing medicinal products in women of childbearing potential	Following safety referral	OTS	Drug Utilisation Study	IPCI (Nl) **SIDIAP (Spain)** CPRD Gold (UK) IQVIA LPD (Be) IQVIA DA (Ge) HDSF (Fi)
DARWIN EU® - **DUS of Antibiotics** in the 'Watch' category of the WHO AWaRe classification of antibiotics for evaluation and monitoring of use	PRAC/CHMP	OTS	Drug Utilisation Study	IPCI (Nl) CHUBX (France) **SIDIAP (Spain)** IMASIS (Spain) IQVIA DA (Ge) CPRD Gold (UK)
DARWIN EU® - **Background rates** of serious adverse events to contextualise safety assessments in clinical trials and non-interventional studies in adolescent and adult patients with **severe asthma**.	CHMP	Complex (complex phenotype)	Disease Epidemiology	IPCI (Nl) **SIDIAP (Spain)** IMASIS (Spain) CPRD Gold (UK) Estonian Biobank

STUDIES YEAR 2 (1)

Study Title	Committees	Study Type	Type of analysis	Data bases
DARWIN EU® - **Multiple myeloma**: patient characterisation, treatments and survival in the period 2012-2022	HTA/Payers	OTS	Disease Epidemiology and Treatment Pattern analysis	IQVIA DA (Ge) **SIDIAP (Spain)** IMASIS (Spain) Estonian Biobank ACI Varha (Fi) CHUBX (France) IKNL (Nl)
DARWIN EU® **Drug Utilisation Study** of **prescription opioids**.	PRAC	OTS	Drug Utilisation Study	Estonian Biobank IPCI (Nl) **SIDIAP (Spain)** CHUBX (France) IQVIA LPD (Be) IQVIA DA (Ge) ACI Varha (Fi)
DARWIN EU® – **EHDS Use Case**: **Natural history** of **coagulopathy in COVID-19 patients** and persons **vaccinated** against SARS-CoV-2 in the context of the OMICRON variant	EC/EHDS	Complex	SIR	IPCI (NL) **SIDIAP (Spain)** IQVIA DA (Ge) Estonian Biobank CPRD GOLD (UK)
DARWIN EU® - Co-prescribing of endothelin receptor antagonists (ERAs) and phosphodiesterate-5 inhibitors (PDE-5is) in **pulmonary arterial hypertension** (PAH)	CHMP	OTS	Disease Epidemiology and Treatment Pattern analysis	CHUBX (France) CPRD GOLD (UK) Estonian Biobank IQVIA DA (Ge)

STUDIES YEAR 2 (2)

Study Title	Committees	Study Type	Type of analysis	Data bases
DARWIN EU® - **Use of take-home naloxone** for opioid overdose treatment	CHMP	OTS	Drug Utilisation Study	IQVIA DA (Ge) IQVIA DA (Be) CPRD Gold (UK) **SIDIAP**
DARWIN EU® **DUS of medicines with prokinetic properties** in children and adults diagnosed with gastroparesis	NCA	OTS	Drug Utilisation Study	IPCI (Nl) CHUBX (France) **SIDIAP (Spain)** IMASIS (Spain) IQVIA DA (Ge) IQVIA LPD (Be) CPRD Gold (UK)
DARWIN EU® - **Effectiveness of COVID-19 vaccines** against severe COVID-19 and post-acute outcomes of SARS-CoV-2 infection	ECDC/ Vaccine monitoring platform	Complex	Comparative Effectiveness	CPRD Gold IPCI **SIDIAP**

https://www.darwin-eu.org/index.php/studies

¡Muchas gracias por vuestra atención!